

Referee report on Florian Landis, "Consumption based responsibility for global warming and the distribution of impacts of Swiss climate policy"

The author combines the Swiss Household budget survey data of 2015-2017, the last full set available today, with third-party data associating a greenhouse gas footprint to every spending item in that dataset to compute the greenhouse gas footprint of households. This is the "consumption based responsibility for global warming" of the title. Next, he looks for regularities in these data, in particular for a relationship between GHG footprint and income. Third, he examines how existing Swiss climate policy could burden the households by income class through its impact on the prices of the spending items. This is "the distribution of impacts of Swiss climate policy" of the title. Last, he simulates a uniform tax on all GHG emissions and checks again how this burden the households by income class, taking into account the uniform refunding of the revenues of the tax.

What is interesting in this paper:

- At least since the yellow vest movement and the rise in energy poverty, policy makers have become aware that they need to know who is most affected by climate policies. To the extent that policies can be fine-tuned, even a carbon or GHG tax, such information can guide the design of complex climate policy. Furthermore, this information can guide the design of social-cushioning measures.
- Even though it is not the first study relating consumption-based GHG footprint with household type, in particular income class, it does it with better data and much greater rigour than earlier studies. One learns a lot on this from this analysis.
- The idea of relating elements of a complex climate policy with expenditure items – how each policy feature could make these items more expensive – and then showing how this affects different household types is very attractive.

From a scientific point of view, this is not an excellent paper:

- The introduction is relatively poor in references. It cites as a general result that lower-income households spend a larger share of their income on energy- and emission-intensive goods than higher income households, but provides only a single reference for this, and it is specific to transport fuels (lines 88-92). I suppose papers derived from the SHEDS survey could be cited. The introduction also claims that "It has been well established that if climate policy is revenue raising, however, revenue can be redistributed to households (in a lump-sum per-capita fashion, e.g.) and that this revenue recycling is more than enough to compensate the regressive direct effects of carbon pricing and produce overall progressive effects" (lines 96-101), but again only one reference in a special case is provided.
- After computing per capita GHG footprints, the author relates them to possible determinants. This is done in a graphical manner, possible determinant by possible determinant. I wonder whether a more systematic multivariate (regression) analysis would not be possible.

- There seems to be an implicit understanding that climate policies aiming at reducing emissions are price policies, or at least that they burden households with the cost of mitigation or the cost of a penalty (tax). For instance, when reporting a key finding of the paper: "Policies that cause emissions reductions for these [luxury] goods (and luxury versions of those goods in particular) can help ensure that high-income households participate in the emissions reduction effort to an appropriate degree." (lines 210-215). In effect, Swiss climate policy also uses a lot of subsidies to encourage GHG emission reduction, in particular for reducing emissions from heating energy or for switching to electric cars. Such policies might benefit high-income households disproportionately.
- The whole discussion in the first part of section 4.2 ("Pricing CO₂ emissions in Switzerland") of the possible distributional impacts of existing Swiss climate policy refers to CO₂, while the households' footprint was computed in GWP, i.e., related GHG emissions. This is probably why the discussion remains qualitative or refers to other publications, instead of using the results of the earlier chapters.
- The second part of section 4.2 ("Pricing ghg emissions globally") does use the per-capita GHG footprints computed in chapter 3, but it does so only for a uniform GHG tax, when the previous subsection on the existing climate policy shows that there can very well be differentiation by expenditure category. Why not imagine a GHG tax that would leave out or treat differently some sectors or goods, e.g., agriculture/food? In the paper, the predictable regressive effect of the uniform GHG tax is offset through uniform per-capita redistribution of its revenues. I think that the expenditure patterns identified in chapter 3 could have been put to better use.
- The concluding remarks mention actually quite little of the lessons learned from the GHG footprint analysis. Instead, it states results that were not directly shown with the data in the paper and that are not actual contributions of this paper. E.g.: "where luxury goods are emissions-intensive, climate policy that focuses on these goods can ensure that high-income households make a relatively bigger effort to reduce emissions than low-income ones" (lines 1753-1758), or "The current design of the CO₂-levy recycles two thirds of the revenue to households in a per-capita lump-sum fashion. This makes the CO₂-levy overall progressive."

Minor points:

- Figure 1 is strange. It compares spending per capita, which is household spending divided by household size, with permanent income per capita, which is measured as household spending divided by the OECD equivalent household size. I am not sure what we learn from this comparison.
- I was intrigued by the lower figures in Fig. 8 and Fig. 9, which show the GWP intensity of different spending items per income class. How can the GWP intensity of one CHF of heating fuel, for instance, vary by income class? In effect, this is not GWP intensity in the classic sense of GHG emissions per monetary unit of spending. It is the contribution of the different spending items to each household's total GWP or GHG footprint. In other words, it is not GWP intensity of one CHF of heating fuel, but GWP of heating fuel relative to total

spending of a household. I wonder whether another terminology could eliminate this confusion, one that involves the term "share" or "contribution".

- When discussing the climate impact of air travel, a RFI = 2 is used, citing a 2019 paper (lines 1200-1203). I would suggest using the RFI = 3 recommended by Neu, U. (2021). The impact of emissions from aviation on the climate – 2nd edition. Swiss Academies Communications 16(3), <https://scnat.ch/en/id/cSx4y>.
- When referring to the appendices, the word "appendix" got lost. E.g., "a more detailed discussion of this can be found in C".

Recommendations for shortening the paper:

- The last sub-section of 4.2 ("Additional considerations") can be cut. It discusses the possibility of households to adjust their consumption to a CO₂ tax, when the paper is about GHG emissions and, by necessity, ignores consumers' response to changing relative prices.
- There are many figures, of which a good number can be dispensed with because they provide little information (Figure 1) or could be replaced by a sentence (e.g., the lower figures of Fig. 4: per capita GWP ranges between a and b for renters, with a median of c, while it ranges between de and e for owner-occupants with a median of f).

Final recommendations

- The paper could be improved by using a proper statistical analysis of the determinants of personal GHG footprints, or there should be an explanation of why such an analysis is not possible.
- The paper could be improved by using the detailed GHG footprint data by income group to simulate more sophisticated climate policies than just a uniform GHG tax.
- The paper could be shortened as indicated above.
- When this is done, I believe that it is strong and interesting enough to be published in SJES.